



THE KING'S SCHOOL

2006
Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value
- Start a new booklet for each Question
- Put your Student Number and the Question Number on the front of each booklet

Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

(a) Differentiate $y = \log_e (\cos x)$ expressing your answer in simplest form. **2**

(b) Evaluate:

(i) $\int_2^3 \frac{2x}{\sqrt{x^2 - 4}} dx$ using the substitution $u = x^2 - 4$. **3**

(ii) $\int_{\pi}^{\frac{4\pi}{3}} \sin x \cos x dx$ **3**

(c) Solve the following inequality for x , graphing the solution on a number line

$$\frac{1}{x+2} < 3$$
2

(d) Determine the acute angle between the straight lines whose equations are

$$x - y + 1 = 0 \text{ and } 2y = x + 1$$
2

End of Question 1

- (a) Consider the function $y = 2\cos^{-1} \frac{x}{3}$.
- (i) Sketch the graph of this function clearly showing the domain and range. **2**
- (ii) Find the angle, θ , that the tangent to the curve $y = 2\cos^{-1} \frac{x}{3}$ at $x = 0$ makes with the positive direction of the x -axis. **3**
- (b) Find the volume of the solid of revolution formed when the curve $y = \sin x$ is rotated around the x -axis between the lines $x = 0$ and $x = \frac{\pi}{4}$. **3**
- (c) Find all values of θ for which $2\sin\theta - \sqrt{2} = 0$. **2**
- (d) Find the point P which divides the interval joint A(-3, 5) and B(7,10) externally in the ratio 2 : 7. **2**

End of Question 2

- (a) A spherical balloon is expanding so that its volume $V \text{ mm}^3$ increases at a constant rate of 72 mm^3 per second.

What is the rate of increase of its surface area $A \text{ mm}^2$, when the radius is 12 mm ? **4**

- (b) Factorise $x^3 - 3x^2 - 10x + 24$, given that $x = 2$ is a zero, and hence solve $x^3 + 24 > 3x^2 + 10x$. **4**

- (c) Solve $3\sin x + 2\cos x = 2$, for $0 \leq x \leq 360^\circ$, to the nearest minute. **4**

End of Question 3

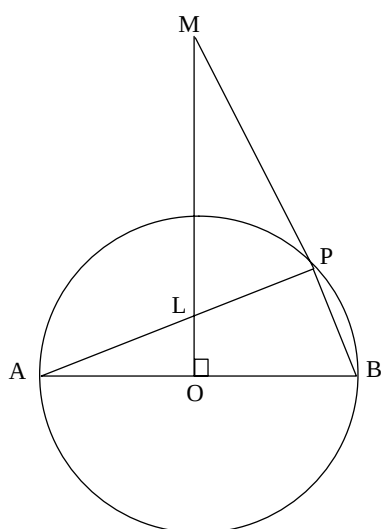
(a) Find the term independent of x in the expansion of $\left(x + \frac{1}{x^2}\right)^9$. 3

(b) Consider the expansion of $(1 + 2x)^n$.

(i) Write down an expression for the coefficient of the term in x^4 . 2

(ii) The ratio of the coefficient of the term in x^4 to that of the term in x^6 is 5 : 8. Find n . 3

(b)



O is the centre of the circle, MPB is a straight line and OLM is perpendicular to AOB as shown. Prove that:

(i) A, O, P, M are concyclic, and 2

(ii) $\angle OPA = \angle OMB$. 2

End of Question 4

- (a) TA and TB are two tangents drawn to a circle from an external point T . A and B are the point of contact of the tangent with the circle.

(i) Draw a neat diagram clearly showing this information.

(ii) Prove that $TA = TB$. 2

- (b) Prove by Mathematical Induction that $7^n - 1$ is divisible by 6 for all positive integers of n . 4

- (c) The elevation of hill at a place A due east of it is 39° , at a place B due south of A , the elevation is 27° .

If the distance from A to B is 500m, find the height of the hill, to the nearest metre. 4

- (d) Show that $\frac{d}{dx}(\log_e 2x) = \frac{d}{dx}(\log_e x)$.

Does this mean that $\log_e 2x = \log_e x$?

Give reasons for your answers. 2

End of Question 5

- (a) (i) Consider the parabola $y = x^2$.

Find the equation of the tangent to the parabola at the point $P(t, t^2)$. 2

- (ii) Show that the line passing through the focus of the parabola and perpendicular to the tangent at P had equation $x = \frac{t}{2}(1 - 4y)$. 2

- (iii) Find the locus of $Q(X, Y)$, the point of intersection of the tangent and the line through the focus perpendicular to the tangent. 2

- (b) A particle A is projected horizontally at 50 m/s from the top of a tower 100m high. At the same instant, another particle B is projected from the bottom of the tower, in the same vertical plane at 100 m/s with elevation 60° .

Prove that the particles will collide and find where they do so. (Use $g = 10\text{ms}^{-2}$.) 6

End of Question 6

Question 7 appears on next page

(a) Solve for $0^\circ \leq \theta \leq 360^\circ$, $\sin x = 3\cos(x + 65^\circ)$. **3**

(b) Prove that the area of a $\triangle ABC$ is $\frac{a^2 \sin B \sin C}{2 \sin A}$. **2**

(c) Given that $y = \frac{x^2 + \lambda}{x + 2}$ and x is real, find:

(i) the set of value(s) of λ for which y can take all but one real value. **1**

(ii) If when $\lambda = 5$, by sketching $y = \frac{x^2 + 5}{x + 2}$, find the range of the function. **3**

(d) Find the values of m for which the line $y = mx$ touches the curve $y = \frac{2x^2 + 1}{2(x + 2)}$. **3**

End of Examination

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$

$$\begin{aligned} 1a) \quad y &= \log_e(\cos x) \\ y' &= \frac{-\sin x}{\cos x} \\ &= -\tan x \end{aligned}$$

$$\begin{aligned} (b) (i) \quad \int_2^3 \frac{2x \, dx}{\sqrt{x^2-4}} \quad & u = x^2 - 4 \\ \frac{du}{dx} &= 2x \\ x=2, u=0 & \\ x=3, u=5 & \end{aligned}$$

$$\begin{aligned} I &= \int_0^5 \frac{2x}{\sqrt{u}} \times \frac{du}{2x} \\ &= \int_0^5 u^{-\frac{1}{2}} \, du \\ &= \left[2u^{\frac{1}{2}} \right]_0^5 \\ &= 2\sqrt{5}. \end{aligned}$$

$$\begin{aligned} (ii) \quad \int_{\pi}^{\frac{4\pi}{3}} \sin x \cos x \, dx \\ &= \int_{\pi}^{\frac{4\pi}{3}} \frac{1}{2} \sin 2x \, dx \\ &= \left[-\frac{1}{4} \cos 2x \right]_{\pi}^{\frac{4\pi}{3}} \\ &= -\frac{1}{4} \cos \frac{8\pi}{3} + \frac{1}{4} \cos 2\pi \\ &= -\frac{1}{4} \cos \frac{2\pi}{3} + \frac{1}{4} \\ &= \frac{1}{8} + \frac{1}{4} \\ &= \frac{3}{8}. \end{aligned}$$

$$\begin{aligned} (c) \quad \frac{1}{x+2} &< 3 \quad x \neq -2 \\ \text{C.P. } x &= -2 \\ 1 &= 3(x+2) \\ x &= -1\frac{2}{3} \end{aligned}$$

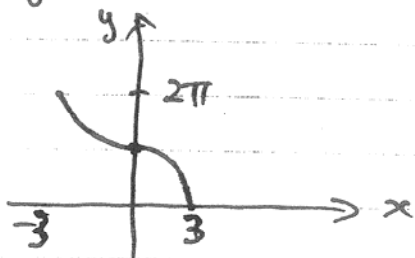
$$\begin{array}{c} \checkmark \qquad \qquad \checkmark \\ -2 \qquad -1\frac{2}{3} \end{array}$$

$$\begin{aligned} \text{test } x=0 \quad \frac{1}{2} &< 3 \text{ true} \\ \therefore x &< -2 \text{ or } x > -1\frac{2}{3}. \end{aligned}$$

$$\begin{aligned} (d) \quad l_1: m_1 &= 1 \\ l_2: m_2 &= \frac{1}{2} \\ \tan \theta &= \left| \frac{1 - \frac{1}{2}}{1 + 1 \times \frac{1}{2}} \right| \\ &= \left| \frac{\frac{1}{2}}{\frac{3}{2}} \right| \\ &= \frac{1}{3} \\ \therefore \theta &= 18^\circ 26' \end{aligned}$$

2a) $y = \cos^{-1} x$ D: $-1 \leq x \leq 1$
 R: $0 \leq y \leq \pi$

(i) $y = 2 \cos^{-1} \frac{x}{3}$



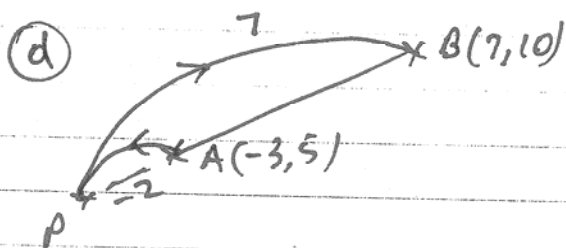
(ii) $y = 2 \cos^{-1} \frac{x}{3}$
 $y' = 2 \times \frac{-1}{\sqrt{1 - \frac{x^2}{9}}} \times \frac{1}{3}$
 $= \frac{-2}{\sqrt{9 - x^2}}$

at $x = 0$ $y' = -\frac{2}{3}$
 $\therefore \tan \theta = -\frac{2}{3}$
 $\theta = 146^\circ 19'$

(b) $V = \pi \int_0^{\pi/4} \sin^2 x \cdot dx$
 $= \frac{\pi}{2} \int_0^{\pi/4} (1 - \cos 2x) dx$
 $= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4}$
 $= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} - (0) \right]$
 $= \frac{\pi}{2} \times \left(\frac{\pi}{4} - \frac{1}{2} \right)$
 $= \frac{\pi(\pi - 2)}{8} \text{ u}^3$

(c) $2 \sin \theta - \sqrt{2} = 0$
 $\sin \theta = \frac{\sqrt{2}}{2}$
 $= \frac{1}{\sqrt{2}}$

$\theta = n\pi + (-1)^n \frac{\pi}{4}$



$x = \frac{-14 - 21}{7 - 2}$ $y = \frac{-20 + 35}{7 - 2}$
 $= -\frac{35}{5}$ $= \frac{15}{5}$
 $= -7$ $= 3$

P(-7, 3)

Q3 @ $\frac{dV}{dt} = 72$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2$$

$$\frac{dr}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$= \frac{1}{4\pi \times 12^2} \times 72$$

$$= \frac{1}{8\pi}$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 8\pi \times 12 \times \frac{1}{8\pi}$$

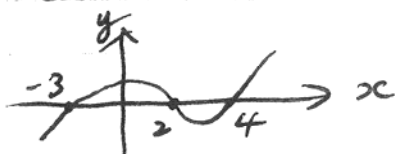
$$= 12 \text{ mm}^2/\text{sec.}$$

(b) $x-2 \overline{) \begin{array}{r} x^3 - x^2 - 12 \\ x^3 - 3x^2 - 10x + 24 \\ \hline -x^2 - 10x \\ -x^2 + 20x \\ \hline -12x + 24 \\ -12x + 24 \\ \hline \end{array}}$

$$x^3 - 3x^2 - 10x + 24 = (x-2)(x-4)(x+3)$$

$$x^3 + 24 > 3x^2 + 10x$$

$$x^3 - 3x^2 - 10x + 24 > 0$$



$$-3 < x < 2 \text{ or } x > 4$$

(c) $3\sin x + 2\cos x = 2$

$$3\sin x + 2\cos x$$

$$\equiv R \sin(x+\alpha)$$

$$\equiv R [\sin x \cos \alpha + \cos x \sin \alpha]$$

$$R = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$\cos \alpha = \frac{3}{\sqrt{13}}$$

$$\alpha = 33^\circ 41'$$

$$\therefore \sqrt{13} \sin(x+33^\circ 41') = 2$$

$$\sin(x+33^\circ 41') = \frac{2}{\sqrt{13}}$$

$$\therefore x+33^\circ 41' = 33^\circ 41' \text{ or } 146^\circ 18'$$

$$\text{or } 393^\circ 41'$$

$$\therefore x = 0^\circ, 112^\circ 38' \text{ or } 360^\circ$$

or $3\left(\frac{2t}{1+t^2}\right) + 2\left(\frac{1-t^2}{1+t^2}\right) = 2$

$$6t + 2 - 2t^2 = 2 + 2t^2$$

$$2t^2 - 3t = 0$$

$$t(2t-3) = 0$$

$$t = 0 \text{ or } \frac{3}{2}$$

$$\tan \frac{x}{2} = 0 \text{ or } \frac{3}{2}$$

$$\frac{x}{2} = 0^\circ, 180^\circ \text{ or } 56^\circ 19'$$

$$x = 0^\circ, 360^\circ \text{ or } 112^\circ 38'$$

Q4) @

$$\begin{aligned}\text{General Term} &= {}^9C_r x^r \left(\frac{1}{x^2}\right)^{9-r} \\ &= {}^9C_r \frac{x^r}{x^{18-2r}} \\ &= {}^9C_r \cdot x^{3r-18}\end{aligned}$$

$$\therefore 3r-18 = 0$$

$$r = 6$$

$$\text{Term is } {}^9C_6 = 84.$$

(b) (i) $(2x+1)^n$

$$\begin{aligned}\text{General term} &= {}^nC_r (2x)^r \cdot 1^{n-r} \\ &= {}^nC_r (2x)^r\end{aligned}$$

$$\text{Term in } x^4 \quad r=4$$

$$\therefore \text{coeff} = {}^nC_4 \cdot 2^4$$

(ii) Term in $x^6 \quad r=6$

$$\text{coeff} = {}^nC_4 \cdot 2^6$$

$$\therefore \frac{{}^nC_4 \cdot 2^4}{{}^nC_6 \cdot 2^6} = \frac{5}{8}$$

$$\frac{\frac{n!}{4!(n-4)!}}{\frac{n!}{6!(n-6)!}} \cdot 2^2 = \frac{5}{8}$$

$$\frac{n!}{4!(n-4)!} \times \frac{6!(n-6)!}{n!} = 4 \times \frac{5}{8}$$

$$\frac{6 \times 5}{(n-4)(n-5)} = \frac{5}{2}$$

$$12 = n^2 - 9n + 20$$

$$n^2 - 9n + 8 = 0$$

$$(n-1)(n-8) = 0$$

$$n \neq 1 \therefore n = 8$$

(c) (i) $\angle AOM = 90^\circ$ (data)

$\angle APB = 90^\circ$ (angle in semicircle)

$\angle APM = 90^\circ$ (rt. line)

$\therefore A, O, M, P$ are

$\angle AOM = \angle APM$

(angles in same seg on circle)

(ii)

$\angle OAP = \angle OMP$

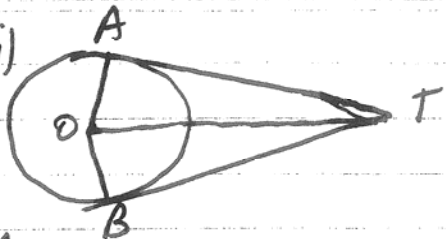
(angles in same seg)

$\angle OAP = \angle OPA$

(base angles of isos Δ)

$\therefore \angle OPA = \angle OMB$

Q5(ii)



In Δ 's OAT & OBT

$OA = OB$ (radius of circle)

OT is common

$\angle OAT = \angle OBT = 90^\circ$

(tang + radius make 90°)

$\therefore \Delta$'s congruent (RHS)

$\therefore TA = TB$

(corresponding sides
of congruent triangles)

⑥ when $n=1$ $7^n - 1$ is divisible by 6

assume true for $n=k$

i.e. $\frac{7^k - 1}{6} = m$ where m is +ve integer

$\therefore 7^k = 6m + 1$

prove true for $n=k+1$

i.e. $\frac{7^{k+1} - 1}{6} = m_1$ (a +ve integer)

$$LHS = \frac{7 \times 7^k - 1}{6}$$

$$= \frac{7 \times (6m + 1) - 1}{6}$$

$$= \frac{42m + 7 - 1}{6}$$

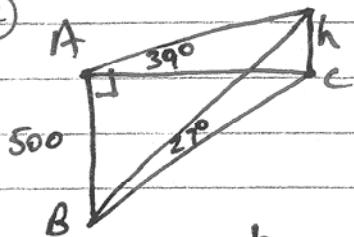
$$= 7m + 1 \text{ (a positive integer)}$$

$\therefore$ if true for $n=k$, it is true for $n=k+1$

since it is true for $n=1$, it is true for $n=2$

and so on for any positive integers

Q5(c)



$$\tan 39^\circ = \frac{h}{AC}$$

$$AC = \frac{h}{\tan 39^\circ}$$

$$\tan 27^\circ = \frac{h}{BC}$$

$$BC = \frac{h}{\tan 27^\circ}$$

$$BC^2 = 500^2 + AC^2$$

$$\frac{h^2}{\tan^2 27^\circ} = 500^2 + \frac{h^2}{\tan^2 39^\circ}$$

$$h^2 \left(\frac{1}{\tan^2 27^\circ} - \frac{1}{\tan^2 39^\circ} \right) = 500^2$$

$$h = \frac{500}{\sqrt{\frac{1}{\tan^2 27^\circ} - \frac{1}{\tan^2 39^\circ}}}$$

$$= 327.8 \dots$$

$$= 328 \text{ m (to nearest metre)}$$

(d) $\frac{d}{dx} (\log_e 2x)$

$$= \frac{2}{2x}$$

$$= \frac{1}{x}$$

$$\frac{d}{dx} (\log_e x)$$

$$= \frac{1}{x}$$

$$\therefore \frac{d}{dx} (\log_e 2x) = \frac{d}{dx} (\log_e x)$$

No - as they differ by a constant

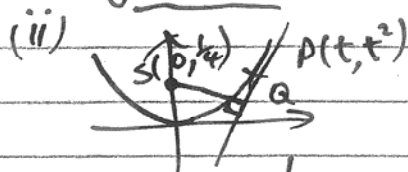
Q 6 (i) $y = x^2$
 $\frac{dy}{dx} = 2x$

at $x = t$
 $\frac{dy}{dx} = 2t$

eqn of tang at P is

$$y - t^2 = 2t(x - t)$$

$$y = 2tx - t^2 \quad (2)$$



$$m_{PQ} = -\frac{1}{2t}$$

eqn of SQ is

$$y - \frac{1}{4} = -\frac{1}{2t}(x - 0)$$

$$-2ty + \frac{1}{2}t = x$$

$$x = \frac{t}{2}(1 - 4y) \quad (1)$$

(iii) solving (1) & (2)

$$y = 2tx - t^2$$

$$y = 2t \times \frac{t}{2}(1 - 4y) - t^2$$

$$y = t^2 - 4t^2y - t^2$$

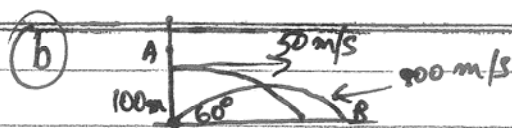
$$y(1 + 4t^2) = 0$$

$$y = 0$$

$$x = \frac{t}{2}$$

$$Q\left(\frac{t}{2}, 0\right)$$

$\therefore$ locus of Q is the x axis.



A: $t=0, x=0, y=100, \dot{x}=50, \dot{y}=0$

B: $t=0, x=0, y=0, \dot{x}=100\cos 60^\circ=50$
 $\dot{y}=100\sin 60^\circ=50\sqrt{3}$

A: $\ddot{x}=0$

$\dot{x}=c_1$

$\dot{x}=50$

$x=50t+c_2$

$c_2=0$

$x=50t$

$\ddot{y}=-10$

$\dot{y}=-10t+c_3$

$c_3=0$

$y=-5t^2+c_4$

$c_4=100$

$y=-5t^2+100$

B: $\ddot{x}=0$

$\dot{x}=c_1$

$c_1=50$

$\dot{x}=50$

$x=50t+c_2$

$c_2=0$

$x=50t$

$\ddot{y}=-10$

$\dot{y}=-10t+c_3$

$c_3=50\sqrt{3}$

$\dot{y}=-10t+50\sqrt{3}$

$y=-5t^2+50\sqrt{3}t+c_4$

$c_4=0$

$y=-5t^2+50\sqrt{3}t$

to collide x & y must be equal at the same time.

$$y = -5t^2 + 100 = -5t^2 + 50\sqrt{3}t$$

$$t = \frac{2}{\sqrt{3}} \text{ sec}$$

if $t = \frac{2}{\sqrt{3}}$ $x = 50 \times \frac{2}{\sqrt{3}} = \frac{100}{\sqrt{3}}$

$$y = -5 \times \left(\frac{2}{\sqrt{3}}\right)^2 + 100$$

$$= 93\frac{1}{3}$$

$\therefore$ particles collide after $\frac{2}{\sqrt{3}}$ sec at $\left(\frac{100}{\sqrt{3}}, 93\frac{1}{3}\right)$

$$Q7 \textcircled{a} \sin x = 3 \cos(x+65^\circ)$$

$$\sin x = 3 \cos x \cdot \cos 65^\circ - 3 \sin x \cdot \sin 65^\circ$$

$$\sin x (1 + 3 \sin 65^\circ) = 3 \cos x \cdot \cos 65^\circ$$

$$\tan x = \frac{3 \cos 65^\circ}{1 + 3 \sin 65^\circ} \quad (\cos x \neq 0, x \neq 90^\circ)$$

$$\therefore x = 18^\circ 50', 198^\circ 50'$$

$$\textcircled{b} A = \frac{1}{2} ab \sin C$$

$$\frac{b}{\sin B} = \frac{a}{\sin A}$$

$$b = \frac{a \sin B}{\sin A}$$

$$\therefore \text{Area} = \frac{1}{2} a \times \frac{a \sin B}{\sin A}$$

$$= \frac{a^2 \sin B}{2 \sin A}$$

$$\textcircled{c} \textcircled{i} \lambda = -4$$

(a)(ii) $y = \frac{x^2+5}{x+2} \quad (x \neq -2)$

$y \neq 0$ as $x^2+5 \neq 0$

$x = 0, y = \frac{5}{2}$

$$y' = \frac{(x+2)2x - (x^2+5) \cdot 1}{(x+2)^2}$$

$$= \frac{2x^2+4x-x^2-5}{(x+2)^2}$$

$$= \frac{x^2+4x-5}{(x+2)^2}$$

for max or min $y' = 0$

$$x^2+4x-5 = 0$$

$$(x+5)(x-1) = 0$$

$$x = -5 \text{ or } 1$$

$x = -5$ LHS $x = -6$ $y' > 0$

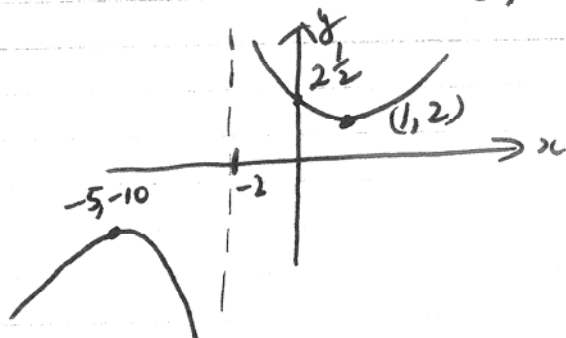
RHS $x = -4$ $y' < 0$

$\therefore$ rel max at $(-5, -10)$

$x = 1$ LHS $x = 0$ $y' < 0$

RHS $x = 2$ $y' > 0$

$\therefore$ rel min at $(1, 2)$



$\therefore y \geq 2 \text{ or } y \leq -10.$

(a) $mx = \frac{2x^2+1}{2x+4}$

$$2mx^2+4mx = 2x^2+1$$

$$x^2(2m-2)+4mx-1=0$$

equal roots

$$16m^2-4(2m-2)x-1=0$$

$$16m^2+8m-8=0$$

$$2m^2+m-1=0$$

$$(2m-1)(m+1)=0$$

$\therefore m = \frac{1}{2} \text{ or } -1.$